

# Terrestrial and aquatic invertebrates in the riverbed of an intermittent river: parallels and contrasts in community organisation

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## SUMMARY

1. Although intermittent rivers (IRs) are coupled aquatic and terrestrial ecosystems, their constituent communities are generally considered separately by ecologists. Yet, both communities are disturbed by alternating flowing and dry phases. IRs are thus good systems in which to explore the parallels and contrasts in community organisation in response to disturbance, and gain understanding in the drivers of these responses.
2. We compared the structure and composition of aquatic and terrestrial communities in the Albarine River, France. We collected aquatic and/or terrestrial invertebrates in the riverbed and adjacent riparian zone at four intermittent reaches and aquatic invertebrates in the riverbed at three perennial reaches over 1 year.
3. A total of 175 taxa were collected at intermittent reaches, including 62 aquatic and 113 terrestrial taxa, contrasting with the low biodiversity generally associated with IRs.
4. Aquatic and terrestrial communities were mainly subsets of those in perennial reaches and in the riparian zone, respectively, indicating that the riverbed of intermittent reaches is strongly linked to perennial reaches and the riparian zone by dispersal.
5. Total taxonomic richness in the riverbed (i.e.  $\gamma$  diversity) of intermittent reaches was mainly due to among-reach variation for terrestrial invertebrates, but to within-reach variation for aquatic invertebrates. For both communities, among-reach dissimilarity was mainly due to spatial turnover rather than community nestedness. Spatial turnover was higher for terrestrial communities than for aquatic communities.
6. Our results indicate that aquatic and terrestrial communities have different organisation patterns in this intermittent river; probably in response to the contrasting spatial arrangements of colonist sources. Heterogeneous sources of terrestrial colonists along the riparian zone compared to more homogeneous sources of aquatic colonists from perennial reaches could explain the relative contribution of nestedness versus turnover.

**Keywords:** dry riverbed, metacommunities, nestedness, riparian zone, species turnover

## Introduction

Intermittent rivers (IRs) are coupled aquatic and terrestrial systems that alternate in space and time between flowing and dry conditions (Larned *et al.*, 2010; Steward *et al.*, 2012; Datry, Larned & Tockner, 2014c). While river

drying is arguably a critical challenge for aquatic communities that has been relatively well-studied over the last decades (Chester & Robson, 2011; Leigh *et al.*, 2015), flow resumption is conversely a pivotal disturbance for terrestrial communities that develop during the dry phase (Corti & Datry, 2012; Rosado, Morais & Tockner,

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2015). Water-dependent ecosystem processes are muted or halted during dry phases. In parallel, although poorly studied, terrestrial ecosystem processes and succession in terrestrial communities begin (von Schiller *et al.*, 2011; Corti, Larned & Datry, 2013) until flow resumption resets succession in aquatic communities and aquatic ecosystem processes. (e.g. Fritz & Dodds, 2004; Febria *et al.*, 2012; Arce *et al.*, 2014). This duality of disturbance makes IRs useful systems in which to explore the parallels and contrasts in community structure and organisation in response to disturbances, and elucidate the drivers of these responses. In addition, IRs are expected to increase in number and length with global change (Döll & Müller Schmied, 2012) and better knowledge of both aquatic and terrestrial communities is essential to predict the changes that will face these ecosystems.

Disturbance can generate species turnover or nestedness among spatially distinct communities (Chase, 2007; Tornés & Ruhí, 2013; Tonkin *et al.*, in press). Species turnover is the replacement of species by others in a community and is generally considered as a response to habitat heterogeneity and the selection of specialist species (Tockner *et al.*, 1999; Hughes *et al.*, 2007). In this case, the diversity of local communities can be low, but compositional variation among communities ( $\beta$  diversity) increases diversity at the regional scale ( $\gamma$  diversity). However, disturbance can also extirpate specialists from the regional pool of species and generate nestedness, so that species-poor communities become subsets of species-rich communities (Ballinger, Lake & MacNally, 2007). Community subsets are common in aquatic communities in IRs (Delucchi, 1989; Davey & Kelly, 2007; Datry *et al.*, 2014b). Aquatic species diversity and density are generally lower in intermittent reaches than perennial reaches (e.g. Feminella, 1996; Arscott *et al.*, 2010; Bogan, Boersma & Lytle, 2013; Datry *et al.*, 2014b), as a consequence of the removal of species that are sensitive to drying (Arscott *et al.*, 2010). Although often overlooked, diverse and abundant terrestrial invertebrate communities have been reported from dry riverbeds (Wishart, 2000; Steward *et al.*, 2011). Terrestrial communities in dry riverbeds may also be subsets of those in adjacent riparian zones, due to the close proximity to this likely source of colonists, and their high mobility (Ballinger, MacNally & Lake, 2005). However, the environmental conditions in dry riverbeds differ substantially from these in riparian zones. Exposed sand and gravel in dry riverbeds are harsh environments (van der Nat *et al.*, 2003). They have low plant diversity and cover compared to riparian zones, especially in arid and temperate climates (Stromberg *et al.*, 2009; Corti & Datry,

2014). Consequently, diel variations in light and temperature are generally greater in dry riverbeds than in adjacent riparian zones, and peak temperatures can be as high as 60 °C (Steward *et al.*, 2011). Because these environmental factors strongly influence terrestrial species (Bates *et al.*, 2007; Lambeets *et al.*, 2008; McCluney & Sabo, 2012), communities from riverbeds and adjacent riparian zones could differ markedly and show strong species turnover (Steward *et al.*, 2011), with some riverbed taxa originating from surrounding upland habitats unable to cope with harsh environmental conditions.

The frequency and duration of drying and rewetting events typically change along river corridors with contraction and expansion cycles (e.g. Larned *et al.*, 2011; Bogan *et al.*, 2013). Aquatic and terrestrial communities may respond differently to these variations. Due to the spatial organisation of the colonist sources, aquatic communities are strongly structured along IRs (e.g. Davey & Kelly, 2007; Arscott *et al.*, 2010). For example, the taxonomic richness of benthic communities decreased along intermittent reaches of increasing drying duration, but not along perennial reaches of an IR in France (Datry, 2012). In addition, few aquatic species are specialists of intermittent reaches (but see Bogan *et al.*, 2013), indicating that communities become species-poorer and simpler with increasing drying frequency and duration. In contrast, such nested patterns may not exist for terrestrial invertebrate communities. Reaches dried for long periods may harbour different species than those more recently dried, due to the progressive change in community structure over time since drying (e.g. germination and growth of ruderal terrestrial vegetation and elimination of water-sensitive species with time) and the intensity of aquatic–terrestrial interactions during the initial phases of drying. In addition, the lateral arrangement of riparian colonist sources and the high sensitivity of riparian species to local conditions such as granulometry and vegetation (Bates *et al.*, 2007) could induce strong species turnover along dry riverbeds. Conversely, the longitudinal dispersal of terrestrial species during sudden rewetting fronts, as well as during dry periods, could decrease species turnover along dry riverbeds (Corti & Datry, 2012; Rosado *et al.*, 2015).

In this study, we described aquatic and terrestrial invertebrate communities alternately colonising the riverbed of a French IR and investigated the parallels and contrasts in their organisation. Over 1 year, we collected (i) benthic aquatic invertebrates in the flowing riverbed at four intermittent and three perennial reaches and (ii) ground-dwelling terrestrial invertebrates in the dry riverbed and the adjacent riparian zone at four intermit-

tent reaches. We predicted aquatic and terrestrial communities at intermittent reaches to be subsets of those in the perennial reaches and the adjacent riparian zone, respectively, due to the substantial role of the riparian zone and the upstream perennial reaches as colonist sources. Second, we predicted that among-reach variation in community structure and composition in the riverbed of the intermittent section would be predominantly driven by nestedness for aquatic communities and by turnover for terrestrial communities, due to the contrasting spatial organisation of colonist sources.

## Methods

### Study reaches

The study was conducted in the Albarine River, an alluvial, gravel-bed IR in eastern France (Fig. 1). The Albarine River drains a 313 km<sup>2</sup> catchment and has a mean annual discharge of 6.88 m<sup>3</sup> s<sup>-1</sup> (Larned *et al.*, 2011). The regional climate is temperate and has wet winters and warm, dry summers. Average annual precipitation is 1000–1200 mm. The downstream section of the Albarine flows across an alluvial plain before joining the Ain River. Along this alluvial plain, the flow regime is intermittent and drying occurs seasonally due to seepage into the underlying aquifer, evapotranspiration and precipitation (Larned *et al.*, 2011). River drying starts at the confluence with the Ain River during low run-off peri-

ods and gradually progresses upstream, leading to a longitudinal gradient in flow intermittence, defined as the percentage of the year a site is dried. The length of the intermittent section is on average 15 km, with the groundwater level often being >10 m below the surface, and hyporheic flow paths are absent, during dry periods (Datry, 2012). Upstream from the intermittent section, the flow regime is perennial (Fig. 1).

Previous studies described in detail the flow regime of the Albarine River, and a statistical model has been calibrated to reconstruct mean daily flows along the river, based on manual measurement of discharge and flow time series from two permanent flow recorders (Larned *et al.*, 2011; Datry, 2012). Mean annual flow intermittence calculated for the last 25 years ranges between 20% at reach D and 55% at reach G. The statistical model was used to generate the flow patterns along the Albarine during the study period and illustrates the alternation of flowing and dry phases in space and time (Fig. 2a).

Four reaches (D–G) in the intermittent section and three reaches (A–C) in the perennial section were sampled for aquatic and/or terrestrial invertebrates (Figs 1 & 2a). Reaches were 2–4 km apart. Reach length was 10× the mean width (18 m) of the active channel across the whole study area to encompass several distinct aquatic and terrestrial habitat types (e.g. gravel and sand bars, forested habitats) (Corti *et al.*, 2013). We defined the active channel as the area of inundated and exposed

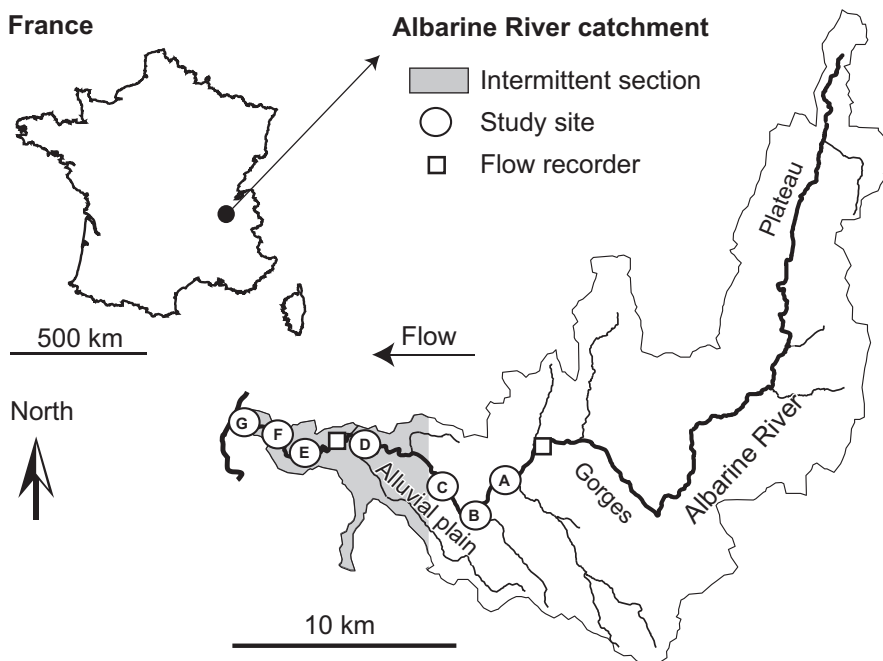
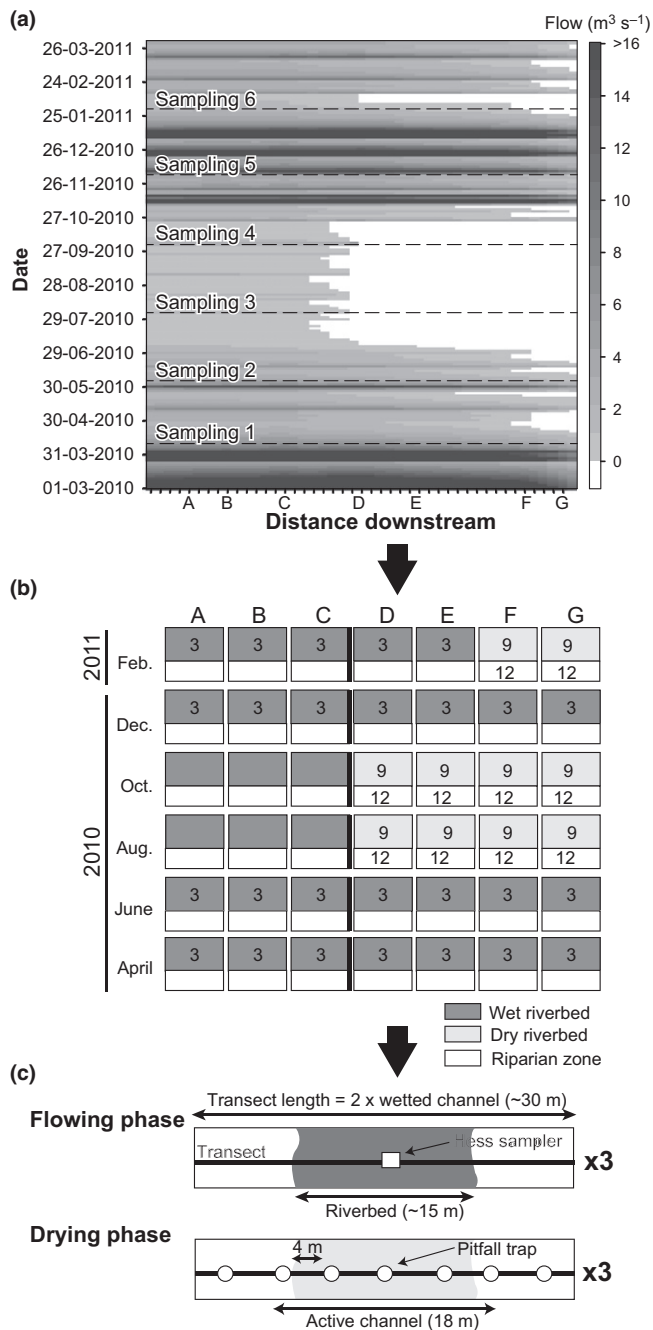


Fig. 1 The Albarine River catchment, showing the intermittent and perennial sections and the seven study reaches (A to G).



**Fig. 2** Diagram of (a) the temporal and spatial pattern of flow intermittence along the Albarine River, (b) the number of benthic and pitfall trap samples taken in each reach during the study period according to flowing and dry phases and (c) the position of pitfall traps and Hess samplers along one transect during a flowing and a dry phase.

bed sediments between established edges of perennial, terrestrial vegetation and abrupt changes in slope (Gregory *et al.*, 1991). The active channel was bordered by a riparian forest dominated by *Populus* sp., *Alnus glutinosa* and *Fraxinus excelsior*.

### Invertebrate sampling

Reaches were sampled at 2-month intervals over 11 months between April 2010 and February 2011 (six sampling dates, Fig. 2a,b). Reaches D to G (intermittent reaches) were sampled for terrestrial and/or aquatic invertebrates depending on dry and flowing phases, and reaches A to C (perennial reaches) were sampled for aquatic invertebrates only (Fig. 2a,b). When an intermittent reach was dry, terrestrial invertebrate communities were sampled in the dry riverbed (DRB) and in the adjacent riparian zone (RZ) (Fig. 2b). DRB was defined as the portion of the active channel over which river flows during base flow conditions of flowing periods (Steward *et al.*, 2011; Corti *et al.*, 2013). RZ was defined as the area extending on both sides of DRB, including gravel bars and vegetated banks (Gregory *et al.*, 1991). On dates when at least one intermittent (INT) reach was sampled for aquatic invertebrates, they were also sampled at the perennial (PER) reaches (Fig. 2b). As a result, terrestrial invertebrates were collected at reaches D to F in August and October 2010 and again in February 2011 at reaches F and G when only these 2 reaches were dried (Fig. 2a, b). Aquatic invertebrates were collected at reaches A to G in April, June and December 2010, and at reaches A to E in February 2011 when only these reaches were flowing.

At each reach, invertebrate sampling took place at three transects separated by 35–45 m, depending on reach length. Transect length was 2× the width of the wetted riverbed (c. 15 m) measured at base flow to include adjacent riparian habitats (Fig. 2c). Ground-dwelling terrestrial invertebrates were sampled using pitfall traps. Although this method generates sampling biases, in particular because catch probabilities increase with species mobility and because diurnal movements cannot be assessed, it is the most efficient method in riparian zones and dry riverbeds (Andersen, 1995; Corti *et al.*, 2013). Seven pitfall traps were installed at each transect and scattered at least 3–4 m apart. This design resulted in three pitfall traps in DRB and four pitfall traps in RZ distributed on either side of the DRB (Fig. 2c). Pitfall traps consisted of 7-cm deep, 8-cm diameter plastic containers set into sediments flush with the riverbed surface. Ethylene glycol (15 mL) was added to each container as a preservative, and a 15-cm diameter plastic plate was held in place above each trap with three wooden sticks to exclude rain. Pitfall traps were deployed for 7 days after which the contents were transferred to plastic bags and stored until laboratory procedures the next day (see below). Although the sampling

design initially included a total of 210 pitfall-trap samples, 15 (7%) were lost due to disturbance by digging mammals (e.g. wild boar). Aquatic invertebrates were collected using a Hess sampler (40-cm diameter, 200- $\mu$ m mesh), transferred to plastic bags and then preserved with 96% ethanol until processed in the laboratory. One sample was collected at each transect.

#### Laboratory procedures

Terrestrial invertebrates from pitfall traps were rinsed with tap water using a 63- $\mu$ m mesh sieve, and the retained material was preserved in 70% ethanol. We identified terrestrial and aquatic invertebrates to the lowest practical taxonomic level, namely species or genus for most Coleoptera, Aranea, Formicidae, insects and molluscs, and family, class or order for other taxa. Species identifications were made by the authors and numerous collaborators (see 'Acknowledgments'), using published keys and reference collections. Aerial terrestrial taxa (e.g. Diptera, Apidae and Lepidoptera) were not included in data analyses as pitfall traps are not an appropriate sampling method for these taxa. Immature spiders (6.2% of individuals) and terrestrial larvae (7.1%) were also removed from the data set because accurate identifications were not possible. Exclusion of immature spiders in data sets generates little bias in taxonomic richness estimates and only a small change in species composition (Sackett, Buddle & Vincent, 2008). Overall, 44% of terrestrial individuals were identified to species, 20% to genus and 36% to family, order or class, and 20% of aquatic individuals were determined to species, 33% to genus and 47% to family, order or class.

#### Data analysis

*Differences in community structure between DRB and RZ and INT and PER.* Taxonomic richness was compared between habitats (i.e. DRB and RZ; INT and PER) by two different procedures, using rarefaction analysis and reach-averaged values. The rarefaction procedure was used to account for the sensitivity of taxonomic richness estimates to the number of samples, taxa occurrence or individuals collected (Gotelli & Colwell, 2001). Rarefaction standardises the sampling effort through random subsampling without replacement of a data set and provides the expected number of taxa in a given number of samples or individuals (Gotelli & Colwell, 2001). Differences among rarefaction curves were compared graphically and were considered statistically significant if the 95% confidence intervals (CI) around rarefaction curves

did not overlap. Reach-averaged taxonomic richness between DRB and RZ and between INT and PER was compared using paired-sample Wilcoxon tests and Wilcoxon tests, respectively, to account for among-reach differences in taxonomic richness. Paired-sample Wilcoxon tests and Wilcoxon tests were also used to test for difference in reach-averaged abundance between DRB and RZ and between INT and PER, respectively.

*Differences in taxonomic composition between DRB and RZ and between INT and PER.* Bray–Curtis dissimilarity index was used to examine differences in terrestrial and aquatic taxonomic composition between DRB and RZ and between INT and PER, respectively. Dissimilarity patterns were compared using non-metric multidimensional scaling (NMDS). A permutational multivariate analysis of variance called Adonis (Oksanen *et al.*, 2014) was used to test for differences in taxonomic composition between DRB and RZ and between INT and PER. Adonis returns a statistic  $R^2$  that measures separation among groups (0 indicates complete mixing and 1 represents full separation), and a  $P$ -value estimated by repeated permutations of the data. We further conducted an indicator species analysis (Indval) (Dufrêne & Legendre, 1997) to assess which taxa were characteristic of DRB and RZ and of INT and PER. Abundance values of DRB, RZ, INT and PER were summed by transect and then averaged by reaches and date to consider between-habitat variability in taxonomic composition only. For NMDS analyses, data were square-root-transformed and double-standardised (Wisconsin method), by standardising taxa by their maxima and then standardising samples by sample totals (Oksanen *et al.*, 2014). Taxa that comprised <0.1% of individuals were removed, to minimise the influence of rare or inadequately sampled taxa.

*Variations in community structure and composition along the intermittent section.* Variation in community structure and composition along the riverbed of the intermittent section (i.e. reaches D, E, F and G) was assessed across two hierarchical levels of spatial scales (i.e. among samples within reaches and among reaches within the Albarine). Invertebrate diversity at each spatial scale was estimated through an additive partitioning of taxonomic richness, with  $\gamma = \alpha + \beta_1 + \beta_2$  (Crist *et al.*, 2003). Gamma-diversity ( $\gamma$ ) refers to the total taxonomic richness,  $\alpha$  refers to the average taxonomic richness in samples, and  $\beta$  refers to compositional variations within each hierarchical level ( $\beta_1$  = among samples within reaches and  $\beta_2$  = among reaches within the Albarine).

The observed diversity components were compared at all hierarchical levels with a null distribution of individuals generated by individual-based randomisations, to calculate expected diversity components and to test the significance of observed patterns of diversity (Crist *et al.*, 2003). The *P*-values estimated by this procedure indicate the probability of obtaining a diversity value as large as (or as small as) it could be expected by chance. Separate diversity partitioning analyses were applied (i) to the entire invertebrate community; (ii) to the community of common taxa (i.e. representing more than 0.3% of individuals) and (iii) to the community of rare taxa (i.e. representing less than 0.1% of individuals), to evaluate whether similar results were obtained with common and rare taxa.

To investigate the underlying mechanisms driving among-reach dissimilarities in terrestrial and aquatic communities,  $\beta$ -reach diversity calculated as Bray–Curtis dissimilarity was partitioned into two components that can produce variation in taxonomic composition: species turnover and nestedness (Baselga, 2010). Turnover (Simpson dissimilarity index) indicates species replacement from one reach to another. Nestedness indicates that communities at reaches with fewer taxa form subsets of communities at reaches with more taxa and reflects a non-random process explaining the differences in the taxa collected due to the loss of taxa (Baselga, 2010). Separate diversity analyses were applied to the entire invertebrate community, to the community of common taxa and to the community of rare taxa.

All statistical analyses were carried out using R software (R Development Core Team, 2014) and functions 'rarefy', 'metaMDS', 'Adonis', 'indval', 'adipart' and 'beta-multi' in the Vegan (Oksanen *et al.*, 2014) and Betapart (Baselga & Orme, 2012) packages. Rarefaction curves were plotted with 95% confidence intervals (CIs) calculated from the conditional standard deviation (Oksanen *et al.*, 2014). For all tests, *P* < 0.05 indicated statistical significance.

## Results

We identified 185 terrestrial taxa and 84 aquatic taxa from 13 646 terrestrial individuals and 82 594 aquatic individuals. (Table S1). A total of 113 taxa were collected in DRB, 142 in RZ, 62 in INT and 79 in PER. Terrestrial communities were dominated by Collembola (57%), Oniscidea (6%), Acarina (6%), Gastropoda (4%) and *Lasius fuliginosus* (Formicidae, 3%). Sixty-one terrestrial taxa (33%) were collected as single occurrences (Table S1). Aquatic communities were dominated by Ostracoda

(24%), Orthoclaudiinae (24%), Chironominae (10%), Oligochaeta (9%) and *Ephemerella* sp. (6%). Seven aquatic taxa (9%, Table S1) were collected as single occurrences.

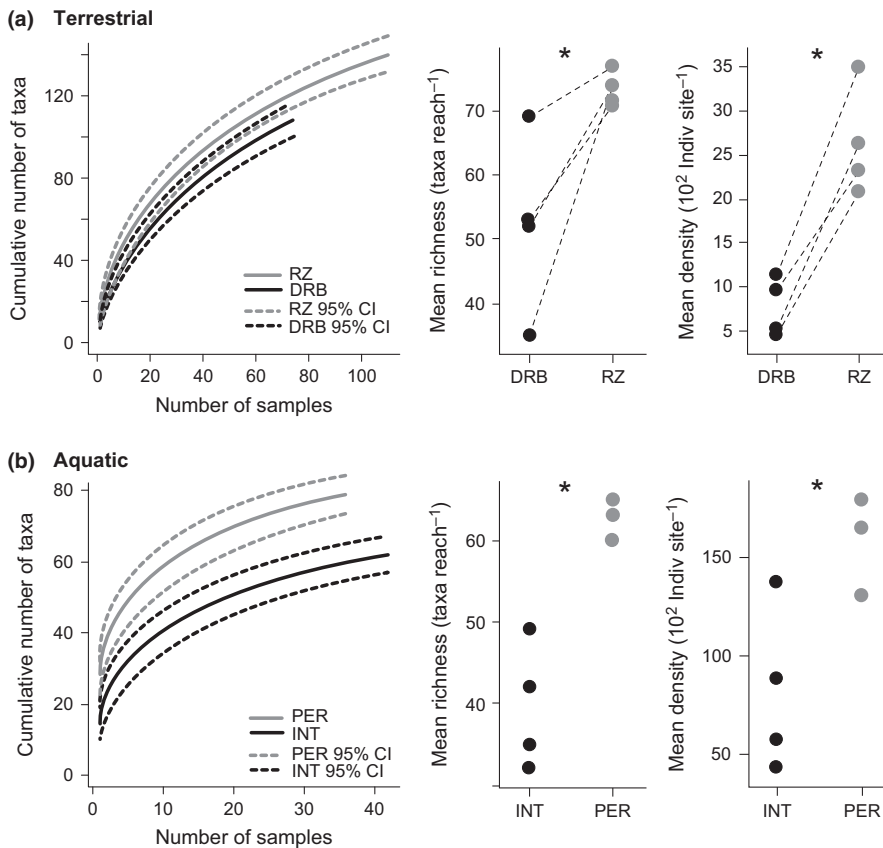
### Differences in community structure between habitats

Terrestrial taxonomic richness and density in DRB were lower than in RZ (Fig. 3a, paired-sample Wilcoxon test, *P* < 0.01), and aquatic taxonomic richness and density in INT were lower than in PER (Fig. 3b, Wilcoxon test, *P* < 0.05). For an identical and maximum number of samples collected (79), 113 terrestrial taxa were collected in DRB whereas 125 taxa were collected in RZ. Similarly, for an identical and maximum number of samples collected (37), 59 aquatic taxa were collected at INT, while 79 taxa were collected at PER. Reach-averaged taxonomic richness was 2.0 times lower in DRB (mean  $\pm$  SD =  $26 \pm 15$  taxa reach<sup>-1</sup>) than in RZ ( $43 \pm 11$ ). Reach-averaged taxonomic richness was 1.8 times lower in INT ( $22 \pm 5$ ) than in PER ( $39 \pm 6$ ). Similarly, reach-averaged terrestrial density was 4.3 times lower in DRB ( $772 \pm 334$  individuals reach<sup>-1</sup>) than in RZ ( $2639 \pm 629$ ), and reach-averaged aquatic density was 2 times lower in INT ( $8100 \pm 4204$ ) than in PER ( $15865 \pm 2556$ ).

### Differences in taxonomic composition between habitats

The composition of terrestrial invertebrate communities differed between DRB and RZ (Fig. 4a). The Adonis analysis revealed a statistically significant separation of DRB and RZ communities based on their taxonomic composition ( $R^2 = 0.67$ , *P* = 0.029). However, only 43 taxa (23%) representing 1% of terrestrial individuals were collected only in DRB, among which 64% were singletons (taxa collected once) (Table 1). There were 72 terrestrial taxa (39%), representing 3% of terrestrial individuals collected only in RZ, among which 56% were singletons (Table 1). There were 70 terrestrial taxa (38%) shared between DRB and RZ. Compositional difference between the dry riverbed and the riparian zone was mostly due to higher relative abundances or occurrences of 15 taxa in RZ than in DRB, mainly *Pseudoscorpiones*, *Myrmecina graminicola* (Formicidae), *Temnothorax* sp. (Araneae), *Nargus* sp. (Coleoptera) and Oniscidea (Table 2). *Chaetocnema* sp. (Coleoptera), *Oedothorax agrestis* (Araneae) and *Oedothorax apicatus* (Araneae) were the only three taxa characteristic of DRB (Table 2).

The composition of aquatic invertebrate communities did not differ between INT and PER (Fig. 4b,  $R^2 = 0.38$ , *P* = 0.093, Adonis). Only five taxa (6%) representing



**Fig. 3** Rarefaction curves (a) for terrestrial invertebrates in DRB (dry riverbed) and RZ (riparian zone) and (b) for aquatic invertebrates in INT (intermittent reaches) and PER (perennial reaches). For mean taxonomic richness and density, each point represents a sampling reach. \* indicates statistical differences based on Wilcoxon tests. Dashed lines link DRB and RZ from the same reach.

0.1% of aquatic individuals were collected only in INT, among which 20% were singletons (Table 1). There were 22 aquatic taxa (26%), representing 0.4% of aquatic individuals collected only in PER, among which 55% were singletons (Table 1). There were 57 aquatic taxa (68%) shared between INT and PER. Taxonomic composition at PER was characterised by higher relative abundances or occurrences of nine taxa compared to INT, mainly *Limnius* sp., *Radix* sp., *Temnothorax* sp., *Hydroptila* sp. and *Riolus* sp. (Table 2). *Niphargus* sp. and Cyclopoida were the only two taxa characteristic of INT (Table 2).

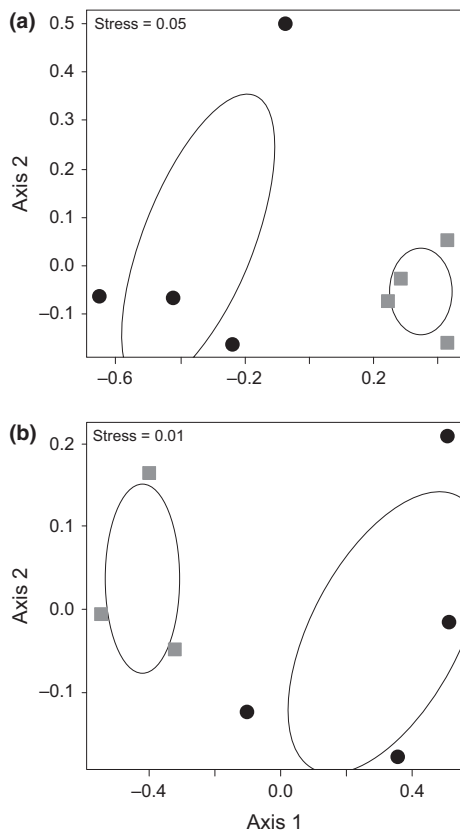
#### Variation in community structure and composition along the intermittent section

The additive partitioning of taxonomic richness showed that except for common aquatic taxa,  $\beta$ -diversity was always larger than expected by chance, indicating that within-reach and among-reach differences in community composition contributed significantly to the total terrestrial and aquatic taxonomic richness (Fig. 5). Total terrestrial taxonomic richness was mainly due to among-reach variation (54%), and total aquatic taxonomic richness was mainly due to among-sample variation (64%).

$\beta_2$  for all, rare and common terrestrial taxa accounted for 54%, 70% and 29% of the total terrestrial taxonomic richness, respectively, whereas  $\beta_2$  for all, rare and common aquatic taxa accounted for 36%, 55% and 6% of the total aquatic taxonomic richness, respectively. In addition, among-reach differences in terrestrial community composition for all, rare and common terrestrial taxa were mostly due to species turnover rather than nestedness (species loss from one reach to another) (Fig. 6). Species turnover for all and rare aquatic taxa also dominated over nestedness but contributed less than for the terrestrial community to among-reach differences in community composition (66% and 75%, respectively). Nestedness dominated (60%) over species turnover for common aquatic taxa.

#### Discussion

Diverse aquatic and terrestrial communities coexist in the Albarine River. However, intermittent reaches harboured lower taxonomic richness and density than perennial reaches and the riparian zone, as well as few specialist species, indicating that aquatic and terrestrial riverbed communities were subsets of those in the



**Fig. 4** Two-dimensional NMDS plots showing the scores of samples and invertebrates collected (a) for terrestrial invertebrates in DRB (black circles) and RZ (grey squares) and for (b) aquatic invertebrates in INT (black circles) and PER (grey squares). Dispersion ellipses represent standard deviation of reach scores of sampling groups. For clarity, only taxa representing more than 0.1% of individuals are shown.

perennial reaches and the adjacent riparian zone, respectively. Compositional variation of both aquatic and terrestrial communities along the riverbed of the intermittent section was dominated by species turnover (rather than nestedness), yet this turnover contributed

more to variation in taxonomic richness and compositional variation of terrestrial than aquatic communities.

#### *Diverse aquatic and terrestrial communities coexist in the Albarine River*

IRs are often viewed as species-poor compared to perennial rivers (e.g. Storey & Quinn, 2008; García-Roger *et al.*, 2011), but this view is inaccurate when aquatic and terrestrial invertebrates are both taken into account over multiple wet and dry periods. In the Albarine River, 175 aquatic and terrestrial taxa were collected over 1 year in the riverbed of intermittent reaches, greatly exceeding the taxonomic richness in perennial reaches (79 aquatic taxa) and in the riparian zone (142 terrestrial taxa). Thus, in terms of the number of species, the colonisation of dry riverbeds by terrestrial communities can compensate for the loss of aquatic species in intermittent reaches (Lake, 2003; Stromberg *et al.*, 2009; García-Roger *et al.*, 2011). Similar results were reported by Katz, Denslow & Stromberg (2012) for plant species, where communities of the intermittent sections of three IRs in Arizona harboured more species than the perennial sections, because many annual species took advantage of seasonally wet or dry soils in intermittent reaches. However, for invertebrates, at least, this high reach-scale biodiversity may be less surprising when considering adjacent riparian habitats and perennial aquatic habitats at the catchment scale, because communities in the riverbed of the intermittent section were subsets of those in the perennial reaches and the adjacent riparian zone (see below). Yet, nested patterns are not always reported for aquatic and terrestrial communities in IRs (e.g. Steward *et al.*, 2011; Bogan *et al.*, 2013). Increasing descriptions of aquatic and terrestrial biota and their biological traits, such as life cycle adaptations and resistant forms as well as the functional roles these

**Table 1** Summary of the terrestrial and aquatic taxa collected from DRB and RZ and INT and PER

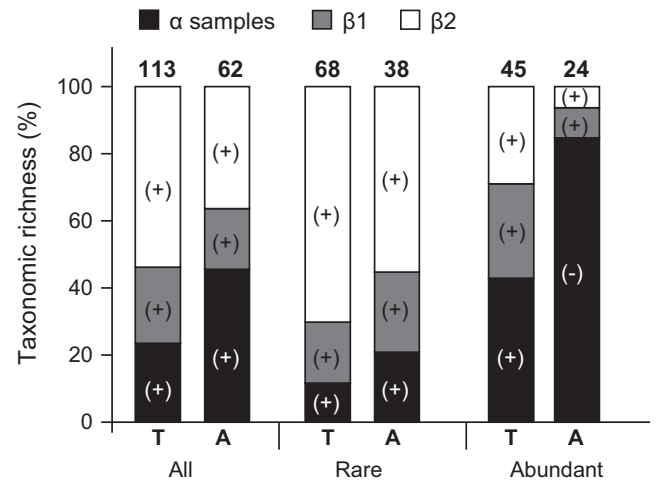
|                            | <i>n</i> | Number of taxa |            |            |         | Number of individuals |            |            |     |
|----------------------------|----------|----------------|------------|------------|---------|-----------------------|------------|------------|-----|
|                            |          | Sum            | Shared (%) | Unique (%) |         | Sum                   | Shared (%) | Unique (%) |     |
|                            |          |                |            | DRB        | RZ      |                       |            | DRB        | RZ  |
| Terrestrial Singletons (%) | 190      | 185 (34)       | 38 (0)     | 23 (64)    | 39 (56) | 13 646                | 96.6       | 0.8        | 2.6 |
|                            |          |                |            | INT        | PER     |                       |            | INT        | PER |
| Aquatic Singletons (%)     | 78       | 84 (15)        | 68 (0)     | 6 (20)     | 26 (55) | 82 594                | 99.5       | 0.1        | 0.4 |

**Table 2** Indicator values (%) for terrestrial taxa in DRB and RZ and for aquatic taxa in INT and PER. Taxa with significantly higher indicator values in DRB and INT are shown in bold type

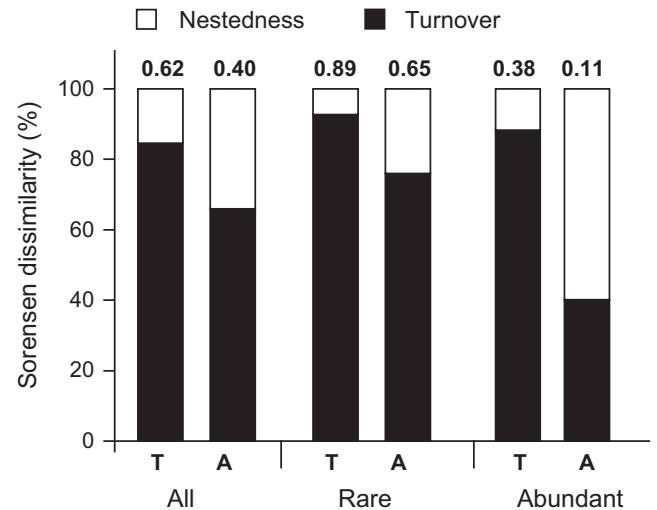
| Terrestrial                                 | Ind values  |             | P            |
|---------------------------------------------|-------------|-------------|--------------|
|                                             | DRB         | RZ          |              |
| Pseudoscorpiones                            | 0.00        | 1.00        | 0.027        |
| <i>Myrmecina graminicola</i> (Formicidae)   | 0.00        | 1.00        | 0.028        |
| <i>Temnothorax</i> sp. (Araneae)            | 0.01        | 0.98        | 0.025        |
| <i>Nargus</i> sp. (Coleoptera)              | 0.02        | 0.97        | 0.021        |
| Oniscidea                                   | 0.04        | 0.96        | 0.030        |
| <i>Tenuiphantes flavipes</i> (Araneae)      | 0.01        | 0.96        | 0.034        |
| Staphylininae sp. (Coleoptera)              | 0.04        | 0.93        | 0.038        |
| Chilopoda                                   | 0.06        | 0.92        | 0.030        |
| Opiliones                                   | 0.09        | 0.91        | 0.031        |
| Diplopoda                                   | 0.08        | 0.90        | 0.034        |
| Siphonaptera                                | 0.08        | 0.89        | 0.024        |
| Aleocharinae (Coleoptera)                   | 0.12        | 0.88        | 0.024        |
| Gastropoda                                  | 0.13        | 0.87        | 0.031        |
| Collembola                                  | 0.19        | 0.81        | 0.023        |
| Acarina                                     | 0.21        | 0.79        | 0.031        |
| <b><i>Chaetocnema</i> sp. (Coleoptera)</b>  | <b>1.00</b> | <b>0.00</b> | <b>0.025</b> |
| <b><i>Oedothorax agrestis</i> (Araneae)</b> | <b>1.00</b> | <b>0.00</b> | <b>0.025</b> |
| <b><i>O. apicatus</i> (Araneae)</b>         | <b>0.99</b> | <b>0.00</b> | <b>0.038</b> |
| Aquatic                                     | INT         | PER         | P            |
| <i>Limnius</i> sp.                          | 0.00        | 1.00        | 0.047        |
| <i>Radix</i> sp.                            | 0.00        | 1.00        | 0.018        |
| <i>Hydroptila</i> sp.                       | 0.00        | 1.00        | 0.030        |
| <i>Riolus</i> sp.                           | 0.00        | 1.00        | 0.030        |
| <i>Odontocerum</i> sp.                      | 0.00        | 1.00        | 0.034        |
| Clinocerinae                                | 0.00        | 1.00        | 0.036        |
| Limoniidae                                  | 0.04        | 0.96        | 0.032        |
| Dolichopodidae                              | 0.04        | 0.96        | 0.031        |
| <i>Rhyacophila</i> sp.                      | 0.07        | 0.90        | 0.049        |
| <b><i>Niphargus</i> sp.</b>                 | <b>1.00</b> | <b>0.00</b> | <b>0.029</b> |
| <b><i>Cyclopoida</i> sp.</b>                | <b>0.82</b> | <b>0.18</b> | <b>0.027</b> |

communities provide, are needed to improve our perception of the biodiversity value associated with IRs.

Remarkably, we found that the dry riverbed of the Albarine River was not species-poor as the high temperatures and the absence of water and vegetation that generally characterise these habitats may suggest. A total of 14 orders, 29 families and 113 terrestrial taxa were collected in the dry riverbed. These communities were dominated by collembolans, two formicids (*Lasius fuliginosus* and *Lasius niger*), acarines, one aranean (*Oedothorax apicatus*) and one coleopteran (*Zorochrus dermestoides*), which represented 76% of the total number of individuals. Collembola, Araneae and Formicidae were also reported by Wishart (2000) and Steward *et al.* (2011) as taxa dominating communities sampled by pitfall traps in dry riverbeds across Australia, Italy and South Africa, suggesting they may be the most abundant ground-dwelling invertebrates in dry riverbeds. Several



**Fig. 5** Relative contribution of  $\alpha$ ,  $\beta 1$  (among samples within reaches) and  $\beta 2$  (among reaches within the Albarine River) to total terrestrial (T) and aquatic (A) taxonomic richness (in bold) in the intermittent section of the Albarine River, for all, rare (<0.1% of abundance) and abundant (>0.3% of abundance) taxa. The observed values were compared to expected values using individual-based randomizations (+ significantly larger, – significantly smaller).



**Fig. 6** Relative contribution of dissimilarity due to nestedness and dissimilarity due to turnover to total dissimilarity among intermittent reaches (in bold) for all, rare (<0.1% of abundance) and abundant (>0.3% of abundance) taxa.

species may migrate diurnally to the riparian zone to avoid the harsh conditions in the dry riverbed and some individuals could use the dry riverbed only at dusk, night and dawn (Steward *et al.*, 2011). Terrestrial richness in the Albarine River was in the range of the data reported by Steward *et al.* (2011), but was one-third of values reported in South Africa by Wishart (2000), although sampling efforts and taxonomic resolution

were similar. Differences in species richness among habitats result from differences in regional species pools, from which species colonise local habitats, but also from differences in local processes, such as biotic interactions and disturbance regime (Belote, Sanders & Jones, 2009). The investigation of local–regional richness relationships across various IRs for aquatic and terrestrial communities could be a promising way to explore the relative contribution of regional and local processes to local diversity.

#### *IR communities are subsets of those in nearby source areas*

As predicted, we found that aquatic and terrestrial ground-dwelling invertebrate communities at intermittent reaches were mainly subsets of those in the perennial reaches and the adjacent riparian zone, respectively, indicating that the riparian zone and the upstream perennial reaches are likely sources of colonists. For aquatic communities, intermittent reaches had (i) lower abundance and taxonomic richness than perennial reaches; (ii) a taxonomic composition similar to perennial reaches and (iii) very few taxa not found in perennial reaches (i.e., indicator species). Similarly, for terrestrial communities, the dry riverbed had (i) lower abundances and taxonomic richness than the riparian zone and (ii) few taxa not also found in the riparian zone. This parallel in community organisation indicates that intermittent riverbed communities are mainly linked to communities in the riparian zone (for terrestrial) and the perennial reaches (for aquatic) by, respectively, lateral and longitudinal species dispersal. The consideration of the combined roles of species dispersal and species sorting by environmental conditions in shaping community structure and composition has led to different metacommunity models (Leibold *et al.*, 2004; Presley, Higgins & Willig, 2010). The mass effect and source–sink paradigms consider species dispersal as the main mechanism generating dissimilarities among different communities while the species-sorting paradigm focuses on the role of local environmental conditions. In the Albarine River, aquatic and terrestrial communities are probably influenced differently by species dispersal and species sorting. For aquatic invertebrates, recent empirical developments suggest that differences in dispersal ability among taxa is the main factor explaining spatial patterns in aquatic communities along intermittent sections (Larned *et al.*, 2010; Robson, Chester & Austin, 2011; Bogan & Boersma, 2012; Datry *et al.*, 2014b). For terrestrial invertebrates, the relatively short distances (i.e. 1–5 m) between the riparian zone and the

dry riverbed compared to the dispersal ability of many terrestrial species (e.g. Datry *et al.*, 2014a) indicate that species dispersal was probably not a limiting factor. Rather, the differences in local environmental conditions between the riparian zone and the dry riverbed and species-sorting processes could dictate terrestrial community organisation. The coexistence of aquatic and terrestrial communities in IRs could offer natural laboratories to gain insight into the respective contribution of local and regional processes for shaping metacommunities.

#### *Contrasts in the spatial organisation of aquatic and terrestrial communities along the intermittent section*

Contrary to our prediction, differences in taxonomic composition among intermittent reaches were mainly driven by species turnover for both aquatic and terrestrial invertebrates. Yet, community nestedness contributed to a higher extent to community dissimilarity for aquatic invertebrates than terrestrial invertebrates and exceeded species turnover for common aquatic taxa. Different spatial dynamics between aquatic and terrestrial communities probably generated this difference in spatial compositional variation. The study of aquatic community patterns along IRs has significantly advanced understanding of ecological effects of flow intermittence (Larned *et al.*, 2011; Bogan & Boersma, 2012). Notably, these previous studies demonstrated that aquatic communities are increasingly simplified with increasing flow intermittence, implying that dispersal from source communities contributes substantially to colonisation and succession dynamics along IRs (Larned, Datry & Robinson, 2007; Arscott *et al.*, 2010; Datry *et al.*, 2014b). In this study, the longitudinal organisation of aquatic communities is suggested by the relatively low beta-diversity among intermittent reaches and the relatively high contribution of nestedness as a driver of among-reach dissimilarity, especially for common taxa. By contrast, beta-diversity among intermittent reaches was high for terrestrial invertebrates and compositional variation among intermittent reaches was largely dominated by species turnover. Longitudinal patterns in aquatic communities along gradient of flow intermittence typically reflect the corridor-like nature of rivers, where river flow constrains the dispersal of aquatic species (Campbell, Lowe & Fagan, 2007). For terrestrial communities, multiple lateral sources of riparian colonists differing in assemblage composition along the intermittent section were probably the cause of compositional dissimilarity among reaches (Corti & Datry, 2014), suggesting that the spatial arrange-

ment of colonist sources is a central aspect of aquatic and terrestrial community organisation along IRs.

Integrating the coexistence of aquatic and terrestrial communities in ecological models (e.g. metacommunities, cross-boundary subsidies) constitutes a future challenge to develop a more comprehensive framework for IRs and better manage their biodiversity in the light of future environmental changes. For instance, although IRs likely are more discrete in time and space than perennial rivers, biotic interactions between aquatic and terrestrial communities during alternating flowing and dry conditions could profoundly affect population dynamics and community assembly and could mitigate the effect of drying disturbances on aquatic and terrestrial communities. From a management perspective, the failure to recognise IRs has led to their serious degradation (Steward *et al.*, 2012; Acuña *et al.*, 2014; Prat *et al.*, 2014). Moreover, IRs are expected to increase in number in the coming decades (Döll & Müller Schmied, 2012) while alterations to IRs are likely to cascade into the perennial networks that include them (Corti & Datry, 2012; Acuña *et al.*, 2014). The contrasting spatial patterns of aquatic and terrestrial communities along the Albarine suggest that management practices aiming to protect aquatic communities, for example through the preservation of colonist sources in perennial reaches, will not necessarily protect terrestrial communities colonising dry riverbeds. Going beyond the historical distinction between aquatic and terrestrial ecology will help to bring the gap between the functioning of the aquatic and terrestrial phases and develop a more holistic view of the ecology of IRs (Datry *et al.*, 2014c).

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## Supporting Information

Additional Supporting Information may be found in the online version of this article:

**Table S1.** Total of individuals (Sum), number of occurrence (Occ), percentage of occurrence (% Occ) and relative abundance (RA, %) of taxa collected in the Albarine River, according to the sampling method (Meth., PF: pit-fall trap, HS: Hess sampler) and the different types of habitat (Hab, B: wet or dry riverbed of intermittent reaches, RZ: riparian zones, PER: perennial reaches).

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